Chapter 5. Transformations, Comparisons and Quadratic Forms

Box–Cox, errors in variables, comparing two lines, multivariate normal and quadratic forms

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5.1 Transformations and the Box–Cox family

When the scatter plot or the residual plot shows curvature or a funnel shape, a transformation of y (or x) can restore linearity, constant variance and normality at once. Common choices are $\log y$, $\sqrt{y}$ and $1/y$. Box and Cox embedded them in one family:

$$y^{(\lambda)} = \begin{cases} \dfrac{y^{\lambda}-1}{\lambda}, & \lambda \neq 0, \\ \log y, & \lambda = 0, \end{cases} \qquad y > 0,$$

which is continuous in λ since $(y^{\lambda}-1)/\lambda \to \log y$. $\lambda = 1$ is (essentially) no transformation, $\lambda = \frac{1}{2}$ the square root and $\lambda = -1$ the reciprocal. The model is

$$y_i^{(\lambda)} = \beta_0 + \beta_1 x_i + \epsilon_i, \qquad \epsilon_i \overset{\text{iid}}{\sim} N(0, \sigma^2),$$

with λ an additional parameter.

Choosing λ. Comparing SSE across λ directly is meaningless, because the scale of $y^{(\lambda)}$ changes with λ. Divide by the Jacobian: with the geometric mean $\dot{y} = (\prod y_i)^{1/n}$, set

$$z_i^{(\lambda)} = \begin{cases} \dfrac{y_i^{\lambda}-1}{\lambda \dot{y}^{\lambda-1}}, & \lambda \neq 0, \\ \dot{y} \log y_i, & \lambda = 0. \end{cases}$$

Then fit $z^{(\lambda)} = \beta_0 + \beta_1 x + \epsilon$ for a grid of λ and choose the λ with the smallest $SSE(\lambda)$.

Proposition 5.1

The profile log-likelihood of λ in the Box–Cox model is $-\frac{n}{2}\log SSE(\lambda) + \text{const}$, where $SSE(\lambda)$ is the residual sum of squares of the regression of $z^{(\lambda)}$. Hence minimizing $SSE(\lambda)$ is maximum likelihood estimation of λ.

Proof. The density of y_i is the normal density of $y_i^{(\lambda)}$ times the Jacobian $y_i^{\lambda-1}$. Maximizing over β and σ^2 for fixed λ gives

$$l_{\max}(\lambda) = -\frac{n}{2}\log\frac{SSE_y(\lambda)}{n} - \frac{n}{2} + (\lambda-1)\sum \log y_i,$$

where $SSE_y(\lambda)$ is the residual sum of squares for $y^{(\lambda)}$. Since $z^{(\lambda)} = y^{(\lambda)}/\dot{y}^{\lambda-1}$, $SSE(\lambda) = SSE_y(\lambda)\dot{y}^{-2(\lambda-1)}$, and $(\lambda-1)\sum \log y_i = n(\lambda-1)\log \dot{y}$. Substituting, $l_{\max}(\lambda) = -\frac{n}{2}\log SSE(\lambda) + \text{const}$. ∎

An approximate 95% confidence set for λ is $\{\lambda : 2\{l_{\max}(\hat{\lambda}) - l_{\max}(\lambda)\} \le \chi^2_{0.05}(1)\}$; in practice one picks a simple value $(0, \frac{1}{2}, 1)$ inside it.

5.2 Measurement error in the explanatory variable

Suppose the true model is $y_i = \beta_0 + \beta_1 x_i^* + \epsilon_i$, but we observe $x_i = x_i^* + \delta_i$, where δ_i has mean 0 and variance σ_δ^2 and is independent of x_i^* and ϵ_i. Treat the x_i^* as random with variance σ_*^2. Then $y_i = \beta_0 + \beta_1 x_i + (\epsilon_i - \beta_1\delta_i)$, and the new error is correlated with the regressor x_i.

Proposition 5.2 (Attenuation)

The least squares slope from regressing y on the observed x satisfies

$$\hat\beta_1 \xrightarrow{p} \frac{\operatorname{Cov}(x,y)}{\operatorname{Var}(x)} = \beta_1 \frac{\sigma_*^2}{\sigma_*^2+\sigma_\delta^2} = \beta_1\Big(1-\frac{\sigma_\delta^2}{\operatorname{Var} x}\Big).$$

Proof. $\hat\beta_1 = \frac{\frac1n\sum(x_i-\bar x)(y_i-\bar y)}{\frac1n\sum(x_i-\bar x)^2}$, a ratio of sample moments that converges to $\operatorname{Cov}(x,y)/\operatorname{Var}(x)$. Here $\operatorname{Cov}(x,y) = \operatorname{Cov}(x^*+\delta, \beta_1 x^*+\epsilon) = \beta_1\sigma_*^2$ and $\operatorname{Var} x = \sigma_*^2+\sigma_\delta^2$. ■

So the slope is biased towards zero (**attenuation** or regression dilution). The bias is small when σ_δ^2 is small relative to the spread of the x's. If the experimenter *sets* the observed value x and the actual value fluctuates around it (Berkson's model, $x^* = x+\delta$), the error is independent of x and least squares remains unbiased.

5.3 Comparing two regression lines

Two groups, $i = 1, 2$, give

$$y_{ij} = \beta_{0i} + \beta_{1i}x_{ij} + \epsilon_{ij}, \qquad \epsilon_{ij} \overset{\text{iid}}{\sim} N(0,\sigma^2), \qquad j = 1,\dots,n_i,$$

with a *common* error variance. This is the **full model**.

Are the lines identical? Test $H_0 : \beta_{01} = \beta_{02},\ \beta_{11} = \beta_{12}$. Under H_0 we have the **reduced model** $y_{ij} = \beta_0 + \beta_1 x_{ij} + \epsilon_{ij}$ for the pooled n_1+n_2 observations.

(1) Fit each group separately: $SSE(F) = SSE_1 + SSE_2$, with $df_F = n_1+n_2-4$.

(2) Fit one line to the pooled data: $SSE(R)$, with $df_R = n_1+n_2-2$.

(3) Compute

$$F_0 = \frac{\{SSE(R)-SSE(F)\}/(df_R-df_F)}{SSE(F)/df_F} = \frac{\{SSE(R)-SSE(F)\}/2}{SSE(F)/(n_1+n_2-4)}.$$

(4) Reject H_0 when $F_0 > F_\alpha(2, n_1+n_2-4)$.

That $F_0 \sim F(2, n_1+n_2-4)$ under H_0 follows from the general theory of Chapters 7 and 8: $SSE(F)/\sigma^2 \sim \chi^2(df_F)$, and $\{SSE(R)-SSE(F)\}/\sigma^2 \sim \chi^2(2)$ independently (a general linear hypothesis with two constraints).

Are the lines parallel? For $H_0 : \beta_{11} = \beta_{12}$, the separate slope estimates are independent, so

$$\operatorname{Var}(\hat\beta_{11}-\hat\beta_{12}) = \sigma^2\Big(\frac{1}{S_{xx,1}}+\frac{1}{S_{xx,2}}\Big), \qquad t_0 = \frac{\hat\beta_{11}-\hat\beta_{12}}{\sqrt{MSE(F)\big(\frac{1}{S_{xx,1}}+\frac{1}{S_{xx,2}}\big)}} \sim t(n_1+n_2-4)$$

under H_0, with $MSE(F) = SSE(F)/(n_1+n_2-4)$. Reject when $|t_0| > t_{\alpha/2}(n_1+n_2-4)$. The same comparisons can be done in a single regression with indicator variables (Chapter 13).

5.4 Multivariate normal distribution

The distribution theory of regression rests on the multivariate normal distribution and quadratic forms.

Definition (Multivariate normal)

$\mathbf{X} \sim N_n(\mu, \Sigma)$, with Σ symmetric nonnegative definite (possibly singular), if any of the following equivalent conditions holds:

(1) $\mathbf{X} \overset{d}{=} A\mathbf{Z} + \mu$ for some $n \times k$ matrix A with $AA' = \Sigma$ and $\mathbf{Z} \sim N_k(\mathbf{0}, I)$;

(2) $\mathbf{X} \overset{d}{=} \Sigma^{1/2}\mathbf{Z} + \mu$ with $\mathbf{Z} \sim N_n(\mathbf{0}, I)$;

(3) $Ee^{\mathbf{t}'\mathbf{X}} = \exp(\mu'\mathbf{t} + \tfrac{1}{2}\mathbf{t}'\Sigma\mathbf{t})$ for all $\mathbf{t} \in \mathbb{R}^n$;

(4) $\mathbf{a}'\mathbf{X} \sim N(\mathbf{a}'\mu, \mathbf{a}'\Sigma\mathbf{a})$ for every $\mathbf{a}$.

The equivalences follow from $Ee^{\mathbf{s}'\mathbf{Z}} = e^{\mathbf{s}'\mathbf{s}/2}$: for instance, (1) gives $Ee^{\mathbf{t}'(A\mathbf{Z}+\mu)} = e^{\mathbf{t}'\mu}e^{\frac{1}{2}\mathbf{t}'AA'\mathbf{t}}$, which is (3). Linear maps preserve normality: $B\mathbf{X} + \mathbf{c} \sim N(B\mu + \mathbf{c}, B\Sigma B')$, and jointly normal vectors are independent iff they are uncorrelated.

5.5 Quadratic forms

Noncentral chi-square: if $X_i \sim N(\mu_i, 1)$ are independent, $\sum_{i=1}^r X_i^2 \sim \chi^2(r, \lambda)$ with noncentrality $\lambda = \frac{1}{2}\sum \mu_i^2$. (Some books use $\sum \mu_i^2$ instead of half of it; we follow the half convention of the lecture.)

Theorem 5.3

If $\mathbf{Z} \sim N(\mathbf{0}, I)$ and A is symmetric with $A^2 = A$, then $\mathbf{Z}'A\mathbf{Z} \sim \chi^2(r)$ with $r = \mathrm{tr}A$. Conversely, if $\mathbf{Z}'A\mathbf{Z}$ is chi-square, then $A^2 = A$.

Proof. Diagonalize $A = P\,\mathrm{diag}(\lambda_i)P'$ with P orthogonal. $A^2 = A$ forces $\lambda_i \in \{0, 1\}$, and the number of ones is $\mathrm{tr}A = r$. With $\mathbf{W} = P'\mathbf{Z} \sim N(\mathbf{0}, I)$, $\mathbf{Z}'A\mathbf{Z} = \sum_i \lambda_i W_i^2$, a sum of r squared independent standard normals. For the converse, compare the mgfs $\prod_i (1 - 2\lambda_i t)^{-1/2}$ and $(1 - 2t)^{-r/2}$: equality of the polynomials $\prod(1 - 2\lambda_i t)$ and $(1 - 2t)^r$ forces each $\lambda_i \in \{0, 1\}$. ∎

Theorem 5.4 (Moments of quadratic forms)

Let $\mathbf{y} \sim N(\mu, V)$ and A symmetric. Then

(a) $E(\mathbf{y}'A\mathbf{y}) = \mathrm{tr}(AV) + \mu'A\mu$ (this needs only the mean and covariance);

(b) $\mathrm{Cov}(\mathbf{y}, \mathbf{y}'A\mathbf{y}) = 2VA\mu$;

(c) $\mathrm{Var}(\mathbf{y}'A\mathbf{y}) = 2\mathrm{tr}\{(AV)^2\} + 4\mu'AVA\mu$.

Proof. (a) $\mathbf{y}'A\mathbf{y} = \mathrm{tr}(A\mathbf{y}\mathbf{y}')$ and $E\mathbf{y}\mathbf{y}' = V + \mu\mu'$. (b), (c) Write $\mathbf{y} = \mu + \mathbf{w}$ with $\mathbf{w} \sim N(\mathbf{0}, V)$: $\mathbf{y}'A\mathbf{y} = \mathbf{w}'A\mathbf{w} + 2\mu'A\mathbf{w} + \mu'A\mu$. Odd moments of $\mathbf{w}$ vanish, so $\mathrm{Cov}(\mathbf{w}, \mathbf{w}'A\mathbf{w}) = 0$ and $\mathrm{Cov}(\mathbf{y}, \mathbf{y}'A\mathbf{y}) = \mathrm{Cov}(\mathbf{w}, 2\mu'A\mathbf{w}) = 2VA\mu$. Likewise $\mathrm{Var}(\mathbf{y}'A\mathbf{y}) = \mathrm{Var}(\mathbf{w}'A\mathbf{w}) + 4\mu'AVA\mu$. Write $\mathbf{w} = V^{1/2}\mathbf{z}$ and diagonalize $V^{1/2}AV^{1/2}$ with eigenvalues λ_i: then $\mathrm{Var}(\mathbf{w}'A\mathbf{w}) = 2\sum \lambda_i^2 = 2\mathrm{tr}\{(AV)^2\}$. ∎

Theorem 5.5

Let $\mathbf{y} \sim N(\mu, V)$ with V nonsingular and A symmetric.

(a) $\mathbf{y}'A\mathbf{y} \sim \chi^2(r(A), \frac{1}{2}\mu'A\mu)$ if and only if AV is idempotent.

(b) In particular, if $\mathbf{y} \sim N(\mu, \sigma^2 I)$, then $\mathbf{y}'A\mathbf{y}/\sigma^2 \sim \chi^2(r(A), \frac{1}{2\sigma^2}\mu'A\mu)$ iff A is idempotent; and $\mathbf{y}'\mathbf{y}/\sigma^2 \sim \chi^2(n, \mu'\mu/(2\sigma^2))$.

Proof of sufficiency in (a). Let $\mathbf{z} = V^{-1/2}\mathbf{y} \sim N(V^{-1/2}\mu, I)$ and $B = V^{1/2}AV^{1/2}$. Then $\mathbf{y}'A\mathbf{y} = \mathbf{z}'B\mathbf{z}$, and $B^2 = V^{1/2}AVAV^{1/2} = V^{1/2}AV^{1/2} = B$ when $AVA = A$, which follows from $(AV)^2 = AV$. Diagonalize $B = P\,\mathrm{diag}(I_r, 0)P'$; with $\mathbf{w} = P'\mathbf{z} \sim N(P'V^{-1/2}\mu, I)$, $\mathbf{z}'B\mathbf{z} = \sum_{i\le r} w_i^2$, a noncentral chi-square with $r = \mathrm{tr}B = r(A)$ degrees of freedom and noncentrality $\frac{1}{2}\sum_{i\le r}(Ew_i)^2 = \frac{1}{2}\mu'V^{-1/2}BV^{-1/2}\mu = \frac{1}{2}\mu'A\mu$. Necessity follows by comparing mgfs as in Theorem 5.3. ■

Theorem 5.6 (Independence)

Let $\mathbf{y} \sim N(\mu, V)$.

(a) $\mathbf{y}'A\mathbf{y}$ and $B\mathbf{y}$ are independent if $BVA = 0$.

(b) $\mathbf{y}'A\mathbf{y}$ and $\mathbf{y}'B\mathbf{y}$ (with A, B symmetric) are independent if $AVB = 0$.

(The conditions are also necessary.)

Proof. We give the proof when A and B are nonnegative definite (for instance idempotent), the case used in regression.

(a) Write $A = LL'$ with L of full column rank. $L'\mathbf{y}$ and $B\mathbf{y}$ are jointly normal with $\mathrm{Cov}(B\mathbf{y}, L'\mathbf{y}) = BVL$. From $BVA = BVLL' = 0$, multiplying by $L(L'L)^{-1}$ on the right gives $BVL = 0$, so $L'\mathbf{y}$ and $B\mathbf{y}$ are independent, and $\mathbf{y}'A\mathbf{y} = \|L'\mathbf{y}\|^2$ is a function of $L'\mathbf{y}$.

(b) The same argument applied to both forms: $\mathrm{Cov}(L_A'\mathbf{y}, L_B'\mathbf{y}) = L_A'VL_B = 0$ when $AVB = 0$. ■

Theorem 5.7 (Cochran)

Let $\mathbf{y} \sim N(\mathbf{0}, I_n)$ and let $A_1, \dots, A_p$ be symmetric with $\sum A_j = I_n$ and $r(A_j) = k_j$. Then the forms $\mathbf{y}'A_j\mathbf{y}$ are independent with $\mathbf{y}'A_j\mathbf{y} \sim \chi^2(k_j)$ if and only if $\sum k_j = n$.

Proof of sufficiency. Since $\sum A_j = I$ and $\sum r(A_j) = n$, the column spaces of the A_j together span $\mathbb{R}^n$ with no overlap: $\mathbb{R}^n = \bigoplus_j \mathrm{col}(A_j)$. For $\mathbf{v} \in \mathrm{col}(A_k)$, $\mathbf{v} = \sum_j A_j\mathbf{v}$ with $A_j\mathbf{v} \in \mathrm{col}(A_j)$; uniqueness of the decomposition gives $A_k\mathbf{v} = \mathbf{v}$ and $A_j\mathbf{v} = \mathbf{0}$ for $j \neq k$. Hence $A_k^2 = A_k$ and $A_jA_k = 0$ $(j \neq k)$. Theorems 5.3 and 5.6 finish the proof. ■

5.6 Expected mean squares in simple regression

Motivation. The F-test compares the regression and error mean squares. Their expectations show why: both estimate σ^2 when $\beta_1 = 0$, while the regression mean square is inflated by $\beta_1^2 S_{xx}$ otherwise.

In the centred model $y_i = \beta_0^* + \beta_1(x_i - \bar{x}) + \epsilon_i$:

Proposition 5.8

$$E(SST) = \beta_1^2 S_{xx} + (n-1)\sigma^2, \qquad E(SSR) = \sigma^2 + \beta_1^2 S_{xx}, \qquad E(SSE) = (n-2)\sigma^2.$$

Proof. $y_i - \bar{y} = \beta_1(x_i - \bar{x}) + (\epsilon_i - \bar{\epsilon})$, so $E(SST) = \beta_1^2 S_{xx} + E\sum(\epsilon_i - \bar{\epsilon})^2 = \beta_1^2 S_{xx} + (n-1)\sigma^2$ (the cross term has mean 0). $E(SSR) = S_{xx}E\hat{\beta}_1^2 = S_{xx}(\sigma^2/S_{xx} + \beta_1^2)$. Subtract. ■

In matrix form, with $J_n = \mathbf{1}\mathbf{1}'$, $SST = \mathbf{y}'(I - J_n/n)\mathbf{y}$; $I - J_n/n$ is idempotent of rank $n-1$, and Theorem 5.4(a) reproduces $E(SST)$.

5.7 Selected exercises

Motivation. The quadratic-form theorems of Section 5.5 are used here to find exact distributions and variances.

Exercise 5.1 (Quadratic forms in a projection)

Let $\mathbf{y} \sim N_n(\mathbf{0}, I_n)$, let X be a fixed $n \times k$ matrix of rank $k < n$, and $H = X(X'X)^{-1}X'$.
(a) Find the distribution of $\mathbf{y}'H\mathbf{y}$.
(b) Find the distribution of $\mathbf{y}'(I - H)\mathbf{y}$.
(c) Find the distribution of $T = \mathbf{y}'H\mathbf{y}/\mathbf{y}'(I - H)\mathbf{y}$.

Solution.

(a) H is symmetric, and $H^2 = X(X'X)^{-1}X'X(X'X)^{-1}X' = H$. Its rank equals its trace, $\operatorname{tr}\{(X'X)^{-1}X'X\} = \operatorname{tr} I_k = k$. By Theorem 5.3, $\mathbf{y}'H\mathbf{y} \sim \chi^2(k)$.

(b) $I - H$ is symmetric with $(I - H)^2 = I - 2H + H^2 = I - H$ and $\operatorname{tr}(I - H) = n - k$. By Theorem 5.3, $\mathbf{y}'(I - H)\mathbf{y} \sim \chi^2(n - k)$.

(c) Since $H(I - H) = H - H^2 = 0$, Theorem 5.6(b) (with $V = I$) shows that the two forms are independent. The denominator is a $\chi^2(n - k)$ variable, positive with probability one, so T is well defined almost surely, and

$$\frac{n - k}{k}\, T = \frac{\mathbf{y}'H\mathbf{y}/k}{\mathbf{y}'(I - H)\mathbf{y}/(n - k)} \sim F(k, n - k).$$

Thus T is distributed as $\frac{k}{n-k}F$ with $F \sim F(k, n - k)$. ■

Exercise 5.2 (Variance of the residual mean square)

(a) Let $y_1, \ldots, y_n$ be i.i.d. $N(\mu, \sigma^2)$ and $S^2 = \sum(y_i - \bar{y})^2/(n - 1)$. Use Theorem 5.4 to show $\operatorname{Var} S^2 = 2\sigma^4/(n - 1)$.
(b) In the regression model $\mathbf{y} \sim N(X\beta, \sigma^2 I_n)$ with X of size $n \times (p+1)$ and full column rank, show $\operatorname{Var}(MSE) = 2\sigma^4/(n - p - 1)$.

Solution [Supplement].

(a) $S^2 = \mathbf{y}'A\mathbf{y}$ with $A = (I - J_n/n)/(n - 1)$, $\mathbf{y} \sim N(\mu\mathbf{1}, \sigma^2 I)$. Since $(I - J_n/n)\mathbf{1} = \mathbf{0}$, $A\mu\mathbf{1} = \mathbf{0}$, so the second term of Theorem 5.4(c) vanishes. Using that $I - J_n/n$ is idempotent with trace $n - 1$,

$$\operatorname{Var} S^2 = 2\operatorname{tr}\{(\sigma^2 A)^2\} = \frac{2\sigma^4}{(n - 1)^2}\operatorname{tr}\Big(I - \frac{J_n}{n}\Big) = \frac{2\sigma^4}{n - 1}.$$

(b) $MSE = \mathbf{y}'B\mathbf{y}$ with $B = (I - H)/(n - p - 1)$. Since $(I - H)X = 0$, $BX\beta = \mathbf{0}$, and since $I - H$ is idempotent with trace $n - p - 1$,

$$\operatorname{Var}(MSE) = 2\sigma^4 \operatorname{tr}(B^2) = \frac{2\sigma^4}{(n - p - 1)^2}\operatorname{tr}(I - H) = \frac{2\sigma^4}{n - p - 1}.$$

(The same follows from $SSE/\sigma^2 \sim \chi^2(n - p - 1)$, whose variance is $2(n - p - 1)$.) The relative standard error of MSE is $\sqrt{2/(n - p - 1)}$, which is large when the residual degrees of freedom are few. ■